

Experimental Determination of Moment of Inertia

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1 Abstract

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

The collected data showed good linearity and values not far from those which could be expected. Slight deviations were however observed and their origin could not be determined to be from either the methods used or the fact that the system deviated slightly from the theoretical system used for the calculations in for instance shape. To determine the origin of the observed deviation, it was concluded that further reasearch would be needed.

2 Introduction

Moment of inertia is a physical property of anything which rotates, characterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate and measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of measuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

To calculate with rotational movement and cannot idealize the mass to a particle as often done in translational calculations, you have to use expressions giving a link between the mass of an object and how it is distributed [2, p. 339]

$I = \text{moment of inertia}$, different geometric forms will give different calculations about its centroid being the center of the distributed mass.

Examples of equations of moment of inertia

Equation for a circular plate

$$I_x = I_y = m \frac{r^2}{2} \quad (1)$$

[2, p. 342]

Equation for a circular ring

$$I_x = I_y = \frac{1}{2} m (r_1^2 + r_2^2) \quad (2)$$

[2, p. 342]

Using this link Newton's second law $\sum F = ma$ analogously can be applied for rotational movement [2, p. 335].

$$\sum \tau = I\alpha \quad (3)$$

τ equals torsion I equals moment of inertia and α the angular acceleration

To calculate torque (τ) it is the distance to where the applied force perpendicular to the rotational center.

$$\tau = Tr \quad (4)$$

[1, p. 171]

T is the applied force in Newton and r is the distance to the point of perpendicular applied force.

Given an expression for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha \quad (5)$$

In the case that the force is generated with a freely falling weight the following substitutions can be made $\sum F = mg - T = ma$. a = the linear acceleration of the falling mass giving the expression.

$$mgr = I\alpha \quad (6)$$

When the centroid of an object and the axes of rotation do not align The Parallel Axis Theorem (Steiner's theorem) can be used to calculate the moment of inertia.

$$I_{total} = I_{cm} + mr^2 \quad (r = \text{the distance between axis}) \quad (7)$$

[1, p. 181]

With the given equations an expression for the experimental moment of inertia can be derived.

$$I = \frac{mgr}{\alpha} - mr^2 \quad (8)$$

For calculations of angular momentum L and rotational energy k_{rot} formulas analogous to linear motion are used.

$$E_{rot} = \frac{1}{2}\omega I \quad (9)$$

$$L = I\omega \quad (10)$$

[1, p. 181]

For error estimation the standard deviation can be calculated to give a sense of how close to the mean value the measurements are distributed. x_i is the experimental value, $\bar{x}$ the mean value and n the amount of data collections

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}} \quad (11)$$

[1, p. 463]

4 Method

Two different experiments were conducted one to experimentally determine the moment of inertia and one to see what happens with angular momentum and rotational energy after an impact with an additional body following along with the rotation.

Both experiments had a common setup with disk, ring, and the weight applying the force in the system. The weight disk and ring were weighed on a laboratory scale and the measurements on the ring and disk were made with a vernier caliper. The experiments were conducted using the PASCO Capstone system with the sensor setup shown in figure 1

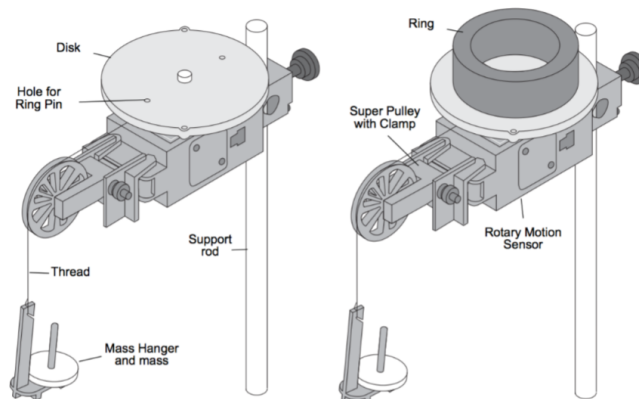


Figure 1: Graphic showing the sensor setup system.

The data was logged with angular velocity and time on the y and x axis in the Pasco system. To have controlled force applied in the system a small weight attached to the disk in the sensor system was used. The weight was released and the PASCO system recorded the event, for the first experiment looking at experimental moment of inertia the disk alone and with an additional ring weight added to the system as in figure 1 was measured five times.

experiment two looking at angular momentum and kinetic energy was conducted in a similar manner but with the ring dropped upon the disk while the weight was still falling. The distance from the approximated ring center to the center of the disk was measured with a vernier caliper.

5 Results

The results in Table 1 show direct measurements of the relevant properties of the experiment setup for calculating the moment of inertia for the ring and disk experiments.

Table 2 and 3 show the data for the experimental calculations for the moment of inertia.

Table 4 and 5 are showing the mean values and standard deviations for the two experiments.

Figures 2 to 5 show the plotted data exported from the CASPECO software for the experiments. Note that the scales on the axes are not the same between the figures.

Table 1: Measurements for ring, disk and attachment point

Entity	Value	Unit
m_{disk}	119,63	g
m_{ring}	457,26	g
m_{weight}	9,66	g
r_{disk}	47,5	mm
$r_{ring\ outside}$	27	mm
$r_{ring\ inside}$	38,5	mm
$r_{attachment\ point}$	23,75	mm

Table 2: Data points for experiment 1

	Disk(rad/s^2)	Disk+ring(rad/s^2)
1	14,0	2,99
2	13,9	3,02
3	14,1	3,02
4	14,1	3,05
5	14,1	3,04
M_α	14,04	3,022
S_α	0,089	0,0238

Table 3: Data points for experiment 2

	Disk(rad/s)	Disk+ring(rad/s)	Displacement(mm)
1	17,554	4,26	6
2	17,606	4,32	8
3	15,093	3,757	4
4	21,271	4,725	9
5	16,153	4,725	5
M_ω	17,53	4,095	6,4
S_ω	2,336	0,498	

Figure 2: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

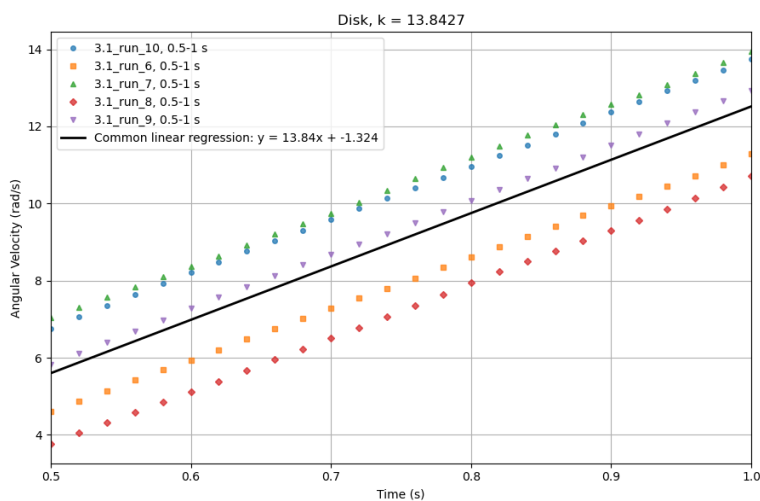


Figure 3: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

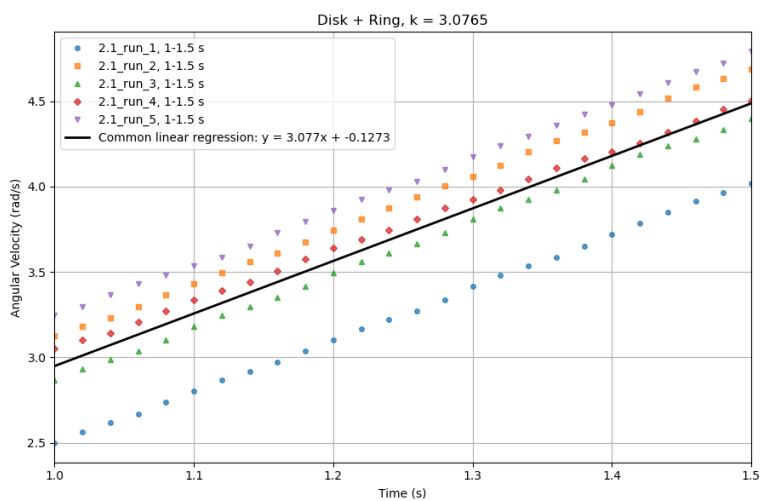


Figure 4: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

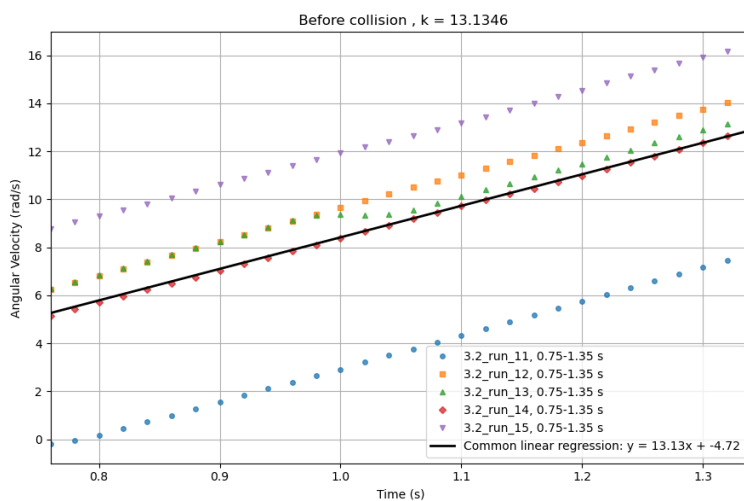


Figure 5: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.

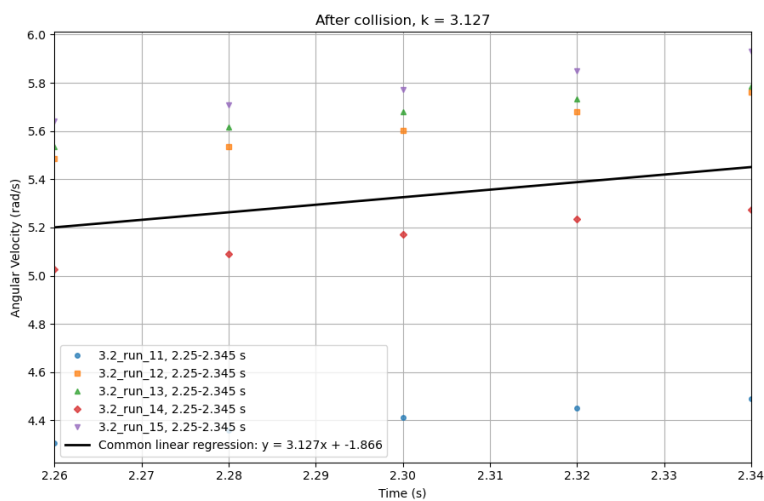


Table 4: Standard deviation calculated according to equation: (11)

Experiment	Mean(α)	Standard deviation
<i>Experimental I_{disk}</i>	14,04	0,089
<i>Experimental $I_{disk+ring}$</i>	3,022	0,0238

Table 5: Standard deviation calculated according to equation: (11)

Experiment	Mean(ω)	Standard deviation
<i>Experimental $L/K_{pre-impact}$</i>	17,53	2,336
<i>Experimental $L/K_{post-impact}$</i>	4,095	0,498

Using equation:(8) the experimental moment of inertia can be calculated only knowing the mass of the falling weight and the radius of the pulling disk.

$$I_{disk} = 1,548kgm^2 \cdot 10^{-4} \quad (12)$$

$$I_{disk+ring} = 7,393kgm^2 \cdot 10^{-4} \quad (13)$$

To calculate the theoretical values for the disk equation: (1) is used and for the disk and the ring it is combined with equation:(2)

$$I_{disk} == 1,349kgm^2 \cdot 10^{-4} \quad (14)$$

$$I_{disk+ring} == 6,388kgm^2 \cdot 10^{-4} \quad (15)$$

Before calculating the kinetic energy and the angular momentum, the angular moment has to be calculated, to calculate with the parallel theorem the mean displacement is used (6,4mm).

$$I_{before} = 1,548kgm^2 \cdot 10^{-4} \quad (16)$$

$$I_{after} = 6,429kgm^2 \cdot 10^{-4} \quad (17)$$

To calculate and compare the kinetic energy before and after the impact equation: (9) is used

$$E_{disk} = 0,001356kgm^2/s^2 \quad (18)$$

$$E_{disk+ring} = 0,001316kgm^2/s^2 \quad (19)$$

To calculate and compare the angular momentum before and after the impact equation: (10) is used

$$L_{disk} = 0,002713 \text{ Kg}m^2/s \quad (20)$$

$$L_{disk+ring} = 0,002632 \text{ kg}m^2/s \quad (21)$$

To compare the before and after impact the percentage change is calculated

$$L_{\%} = \frac{L_e - L_f}{L_f} \cdot 100 = 2,985\% \quad (22)$$

$$K_{\%} = \frac{K_e - K_f}{K_f} \cdot 100 = 2,949\% \quad (23)$$

6 Discussion

All the data show good linearity, indicating that the use of a linear model was indeed appropriate, while forces such as air resistance are exponential in nature they were not of a sufficient magnitude to significantly affect the results. Additionally, none of the data strayed from the theoretical values by completely unexpected amounts such as an order of magnitude. The observed moments of inertia did however also not completely align with the expected values, differing by $1.005 \text{ kgm}^2 \cdot 10^{-4}$ which is not insignificant when the observed value is only $7.393 \text{ kgm}^2 \cdot 10^{-4}$. It cannot, however, be ruled out that this difference could stem from the fact that the samples tested were not perfect cylinders, the system not being completely frictionless, or similar factors. In this case the observed result could be closer to the actual value than the theoretical value, which assumes perfect conditions.

Whether this deviance is to be considered significant depends on the application. If for instance the aim is to ensure a motor in a stationary application has sufficient torque, this method is totally adequate as the resulting moment of inertia is higher than the theoretical one. Dimensioning with these experimental values should therefore always ensure some limited level of safety margin, without further testing, this should however not be relied upon and instead be used in conjunction with additional safety margins.

To find whether the experimentally determined moment of inertia is a better or worse approximation of the actual value than the theoretical one a more accurate model taking into consideration the exact shape of the disk and friction would need to be developed to compare against. Alternatively similar experiments could be run with multiple other shapes to determine the deviation from expected values in those cases, which would give an indication of how much variance comes from the imperfections in measurements versus how much comes from the variance in shape between the actual sample and the one used in the theoretical calculations.

The results of the experiment involving impulse do show the conservation of angular momentum however gave less conclusive results. The angular momentum showed a change of 2.985% after the impact while the kinetic energy showed a change of 2.949%. These values are so similar that they can essentially be regarded as the same. Which does not tell anything about the conservation of angular momentum compared to rotational kinetic energy. It does however show that the change in angular momentum is relatively small and can likely be attributed to losses caused by the system being imperfect.

To better observe this difference and get clearer data that angular momentum is indeed conserved, a more ideal test would need to be conducted such as using a disk with a higher friction surface to minimize sliding between disc and ring. Or dropping the ring from a jig which holds is perfectly centered over the disc.

7 References

References

- [1] Carl Nordling and Jonny Österman. *Physics Handbook for Science and Engineering*. 9th ed. Studentlitteratur, 2020.
- [2] Hugh D. Young and Roger A. Freedman. *University Physics*. 11th ed. Pearson, 2003.